MKTG90011 Marketing Research

THE UNIVERSITY OF MELBOURNE

Seminar 2

Topics for Today

1. **Data**
   - Primary and Secondary Data
   - Quantitative and Qualitative Data
   - Unit of Analysis, Dependence of Observations, Dependent and Independent Variables
   - Measurement Scaling
   - Reliability and Validity
   - The Population and Sampling, Sample Sizes

2. **The Literature Review**

3. **Workshop**

Housekeeping

1. Email: Please include [Marketing Research] or [MKTG90011] in the subject line.
2. What is the difference between a Group of 4 and a Group of 5?
   - Data collection
   - Group of 4: n = 100 (minimum) for quantitative surveys and n = 10 for qualitative interviews (as in the subject guide)
   - Group of 5: n = 125 (minimum) for quantitative surveys and n = 12 for qualitative interviews

3. The first quiz will be available on Thursday at 11:59 pm.
   - 7 Multiple-Choice/True-False questions + 1 Short-Answer question
   - Materials in W1 and W2
   - 90 min

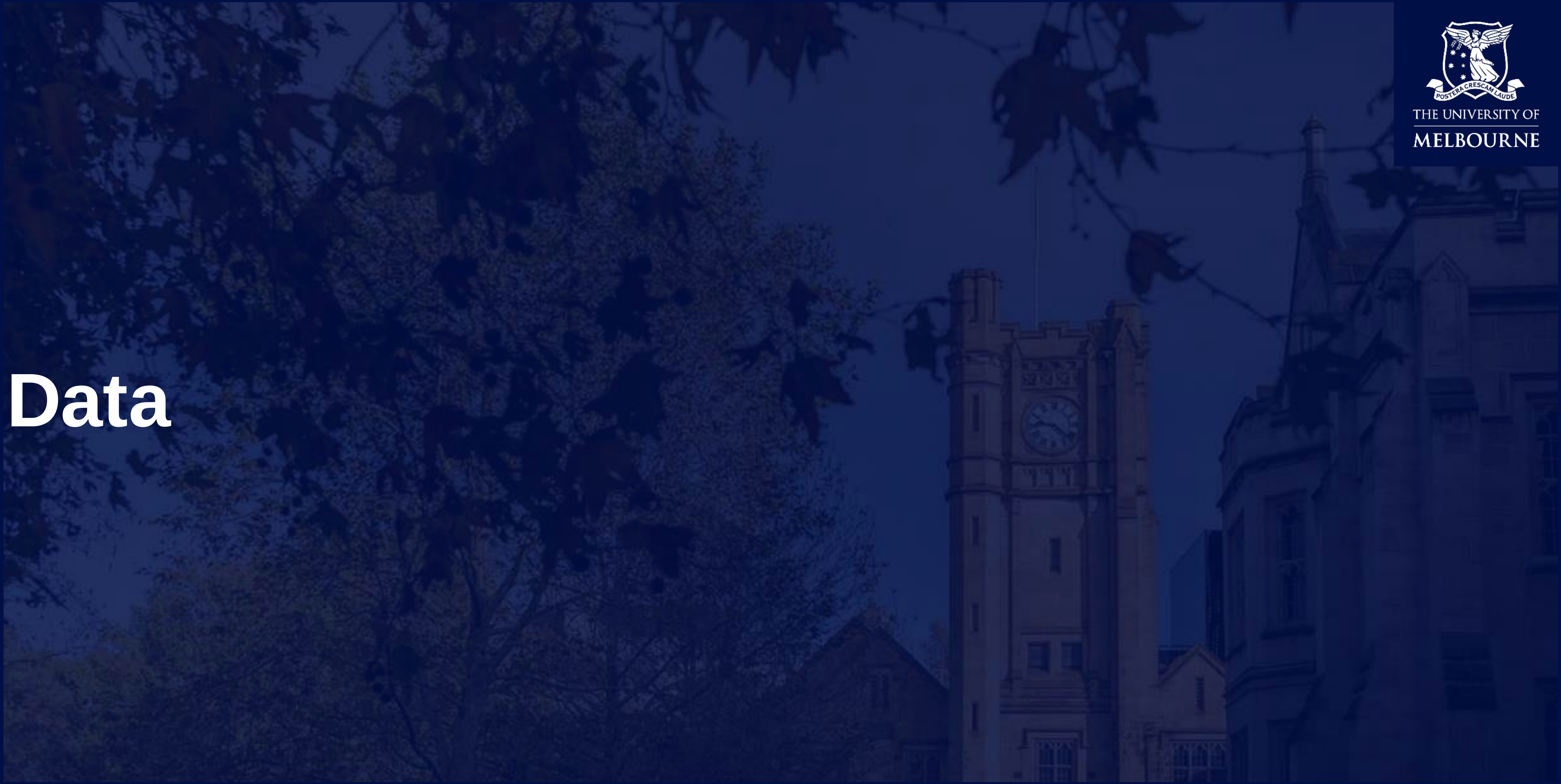
Recap

1. Identify and formulate the problem
2. Determine the research design
3. Design the sample and method of data collection
4. Collect the data
5. Analyze the data
6. Interpret, discuss, and present the findings
7. Follow-up

Ambiguous problems: Exploratory research design

Somewhat defined problems: Descriptive research design

Clearly defined problems: Causal research design



Data

What is data?

Data are facts and statistics collected together for reference or analysis (Oxford dictionary)

	B	C	D	E	F	G	H	I	J	K
ID	Gender	Location	Generation	Weight	Q0	Q1	Q2	Q3	Q4	Q5
2	Male	South America	Generation X	1	0	0	0	1	0	
4	Female	South America	Baby Boomers	1.44	0	1	1	1	0	
5	Female	South America	Generation X	1	1	0	1	1	1	
6	Male	Antarctica	Baby Boomers	1.44	2	0	1	1	0	
9	Female	Europe	Baby Boomers	1.32	1	1	1	1	1	
12	Female	Europe	Baby Boomers	1.56	0	0	0	1	1	
15	Male	North America	Baby Boomers	1.56						
16	Male	Antarctica	Baby Boomers	1.44	1	1	1	0	0	
17	Female	Europe	Baby Boomers	1.32	2	1	1	1	0	
18	Male	North America	Traditionalists	0.595	0	0	1	0	0	
22	Male	South America	Generation X	1.32	0	0	0	1	1	
25	Female	South America	Generation X	1.32	1	1	1	1	1	
26	Female	South America	Millenials	0.765	1	0	1	0	0	
27	Male	Europe	Baby Boomers	1.56	1	1	1	1	0	
29	Male	Europe	Generation X	1						
30	Male	Europe	Baby Boomers	1.32	0	0	0	0	0	
31	Male	Europe	Millenials	0.68	1	0	0	0	0	
33	Male	North America	Generation X	1						
34	Male	North America	Generation X	1.32						
36	Female	North America	Millenials	1	1	0	1	1	1	
37	Female	North America	Millenials	0.765						

Definition of terms (in a data set)

- **Variable:** an attribute whose value can change. A *variable* is the opposite of a *constant*.
- **Constant**: an attribute that does not change its value
- A **case** (or **observation**) consists of all the observed variables that belong to an object such as a customer, a company, or a country
- **Item**: usually refers to a survey question put to a respondent

Construct: a variable that is not directly observable.

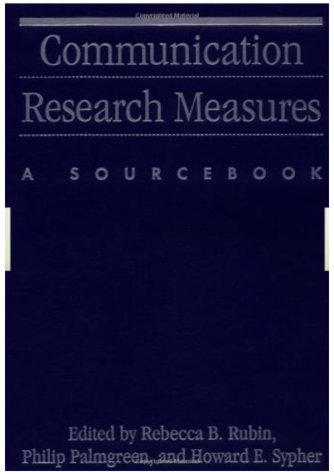
Examples: Happiness; Trust; Satisfaction

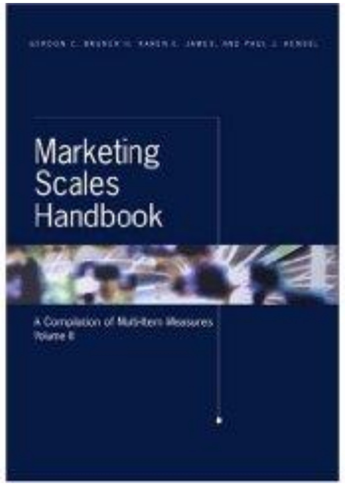
Types of construct:

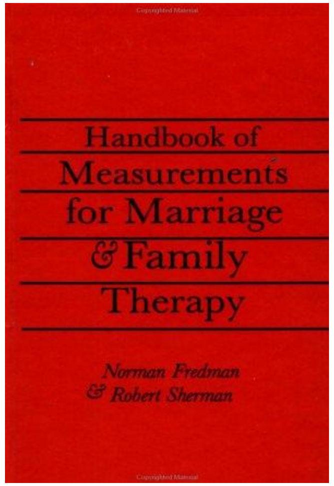
- Reflective vs. Formative
- Single- vs. multi-item

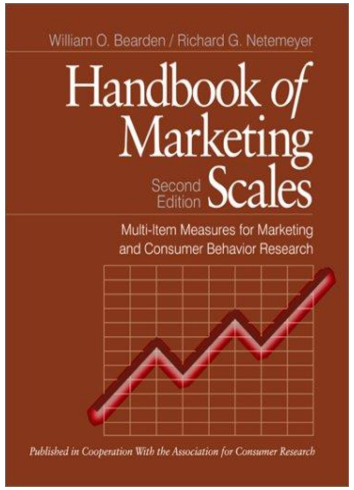
- Developing constructs requires using scale/ index development procedures (more to come in W4)

How to Find Construct Measures









SCALE NAME: **Commitment to the Company (General)**

SCALE DESCRIPTION:

The six item, seven-point Likert-type scale assesses the degree to which a person expresses a willingness to continue a relationship with a company and make some effort to do it if need be.

SCALE ORIGIN:

The scale appears to be original to Aaker, Fournier, and Brasel (2004) although they received inspiration from previous work by others.

RELIABILITY:

Aaker, Fournier, and Brasel (2004) reported alphas ranging from .91 to .93 over three time periods.

VALIDITY:

No examination of the scale's validity was reported by Aaker, Fournier, and Brasel (2004).

COMMENTS:

The company with which the scale was used in the experiment by Aaker, Fournier, and Brasel (2004) was a fictitious website. If the scale is used with other entities then adjustment in some of the items might be necessary, such as #3 (below).

REFERENCES:

Aaker, Jennifer, Susan Fournier, and S. Adam Brasel (2004), "When Good Brands Do Bad," *JCR,* 31 (June), 1-16.

SCALE ITEMS:[1]

1. I am very loyal to __________.
2. I am willing to make small sacrifices in order to keep using __________.
3. I would be willing to postpone my purchases if the __________ site was temporarily unavailable.
4. I would stick with __________ even if it let me down once or twice.
5. I am so happy with __________ that I no longer feel the need to watch out for other alternatives.
6. I am likely to be using __________ one year from now.

[1] The name of the focal company should be placed in the blanks.

Agenda

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2. Primary and Secondary Data
3. Quantitative and Qualitative Data
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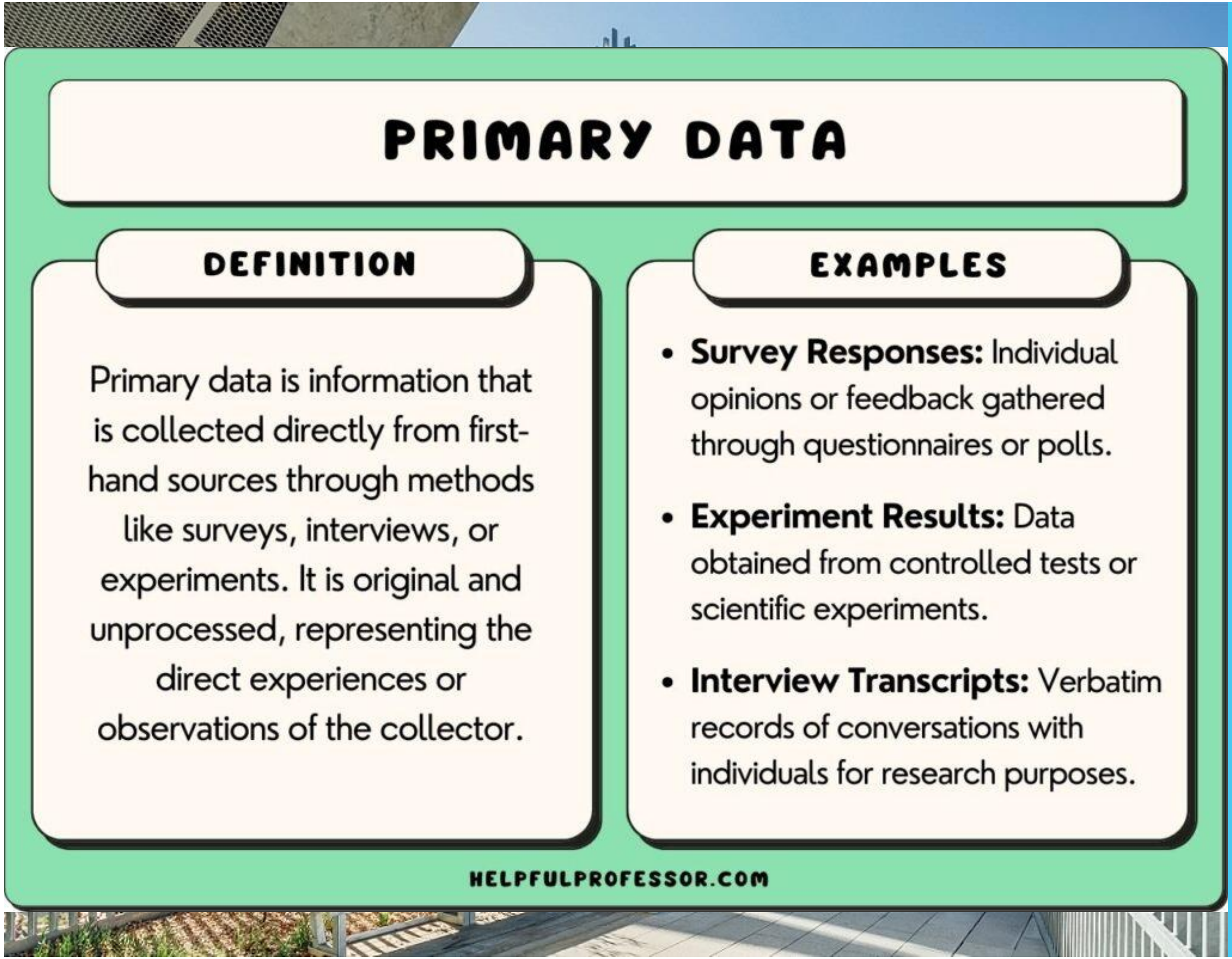
6. Reliability and Validity
7. The Population and Sampling, Sample Sizes

Primary Data



https://helpfulprofessor.com/primary-data-examples/

Secondary Data

Secondary data is collected previously for another purpose

Internal

Examples:
Sales reports

Existing internal research studies

External

Examples:
Industry reports

Data provided by market research firms

Example: https://www.ibisworld.com/australia/industry/vitamin-and-supplement-manufacturing/5417/

Primary vs. Secondary Data

	Secondary data	Primary data
Advantages	• Tends to be cheaper • Sample sizes tend to be greater • Tend to have more authority • Are usually quick to access • Are easier to compare to other research using the same data • Are sometimes more accurate	• Are recent • Are specific for the purpose • Are proprietary
Disadvantages	• May be outdated • May not fully fit the problem • There may be hidden errors in the data • Usually contain only factual data • No control over data collection • May not be reported in the required form	• Are usually more expensive • Take longer to collect

Some examples – internal secondary data

- Sales invoices, accounts receivables reports, sales, social media, customer service reports.
- Customer letters or comment cards, mail-order forms, cash register receipts, credit applications, digital purchase receipts, website and mobile apps, salesperson expense reports, employee exit interviews

Some examples – external secondary data

- IbisWorld (Reports)
- Australian Bureau of Statistics (for market descriptors)
- Reviews (e.g., Google maps)
- Hootsuite (for social media – see video on LMS)
- Hoovers (company information – available via the FBE library)
- LinkedIn Business (To find contacts)
- Major newspapers and periodicals (Bloomberg Businessweek, Forbes, Harvard Business Review, Australian Financial Review)
- Scholarly sources (Google Scholar, EBSCO, ProQuest)
- Government sources (Census data, additional government reports, and publication catalog)
- Commercial sources (Market Line, Passport, Euromonitor)

Secondary Data

Finding Secondary Data using Google:

- Use Google Scholar (http://scholar.google.com) if you are looking for scholarly information such as in academic journals. Often, you can only access the information if you have an organizational, university, or library password.
- With Google Books (http://books.google.com/books) you can search within a catalogue of digital books. Google returns in which books the keywords you have used are found. Google may also provide a text preview.
- Under advanced search, you can tell Google to only return searches from e.g., Excel documents by selecting Excel under file type. Under advanced search you can also tell Google to return searches from universities only by typing in ".edu" in *Search Within Site or Domain* (mainly USA!).
- For news from some time ago (e.g., to see how a new product launch was received by the press) use Google News Archive Search (http://news.google.com/archivesearch).
- Try to use operators to restrict your research. E.g., putting a "–" right before a search word excludes this word from your findings. "+" allows you to search for websites that include that particular word. Putting a sequence of in quotation marks "…" indicates that Google should search for exactly these elements.
- You can use a dedicated search tool to find secondary data such as the Internet Crossroads in Social Science Data (http://www.disc.wisc.edu/newcrossroads/index.asp). These dedicated search tools and lists often work well if you are looking for specialized datasets.

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Quantitative and Qualitative Data

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Quantitative data are presented in values

- Sales in AUD
- Full-time work (0/1)
- Survey items: disagree(0), somewhat disagree (1), neutral (2), somewhat agree (3), agree (4).

Qualitative data may be presented in all other forms

- Observation of customers shopping for products
- Open-ended questions
- Description of holiday experience

Quantitative and Qualitative Research

The difference between qualitative and quantitative market research is not just one of words versus numbers. It has to do with quantification.

- If you know all possible values your variables will take on before collecting data → quantitative research. *Examples:* age (years), temperature (e.g. in Fahrenheit or Celsius)

- If you don't know the possible values, you have qualitative data. *Examples:* how do you feel about Apple's iPad? What made you decide to buy a Mercedes E-class?

- You can turn qualitative data into numbers. For example, you can assign different values to the reasons why people decided to buy a Mercedes E-class.

- The research will remain qualitative however, the data itself are *quantified*.

Typical Methods Associated with Collecting Quantitative and Qualitative Data

Primary data

Quantitative

- Survey (mainly closed questions)
- Experiments

Qualitative

Direct

- Interviews (with experts)
- Focus groups

Indirect

- Observational studies
- Projective methods
- Test markets

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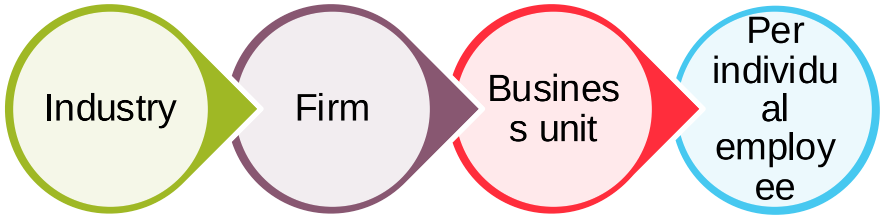
Unit of Analysis

The unit of analysis is the level at which a variable is measured

- Customer satisfaction
- Profitability



Data is best collected at the lowest level

- Date can be aggregated (summed) to higher levels later
- Example: “Cars sold at each individual car dealership” can be aggregated to “Cars sold in region / country XYZ”

Dependence of Observations

Dependence of observations = are observations related (correlated)?

–Imagine we conduct an experiment:

- – We ask participants to rate a type of Cola
- – We show them an advertisement of this type of Cola
- – We ask them to rate the Cola again

–First and second responses will be related: Those who rated the cola negatively (positively) at first, are more likely to continue rating it negatively (positively) next time

Dependent and Independent Variables

Dependent and independent variables are terms relating to the design of a market research study

Dependent variable	• What you try to explain
Independent variable	• What you use to explain it

Do not confuse with ‘does the variable depend on something else.’ Almost all variables depend on something else (e.g., loyalty, price sensitivity etc.)

More to come in W5 (Causal Research)

Measuring Scaling

- Nominal scale (label for the presence / lack of presence of some characteristic; e.g., color of car: blue=1, red=2 etc.)
- Ordinal scale (rank order of objects, customers etc.; e.g., non-user=0, light user=1, and heavy user=2)
- Interval scale (precise differences between scale points; e.g., temperature in Celsius/Fahrenheit)
- Ratio scale (same as interval scale but with a natural origin / zero point; e.g., sales in USD)
- More to come in W4 (Scale Development)

	Label	Order	Differences	Origin is 0
Nominal scale	✓			
Ordinal scale	✓	✓		
Interval scale	✓	✓	✓	
Ratio scale	✓	✓	✓	✓

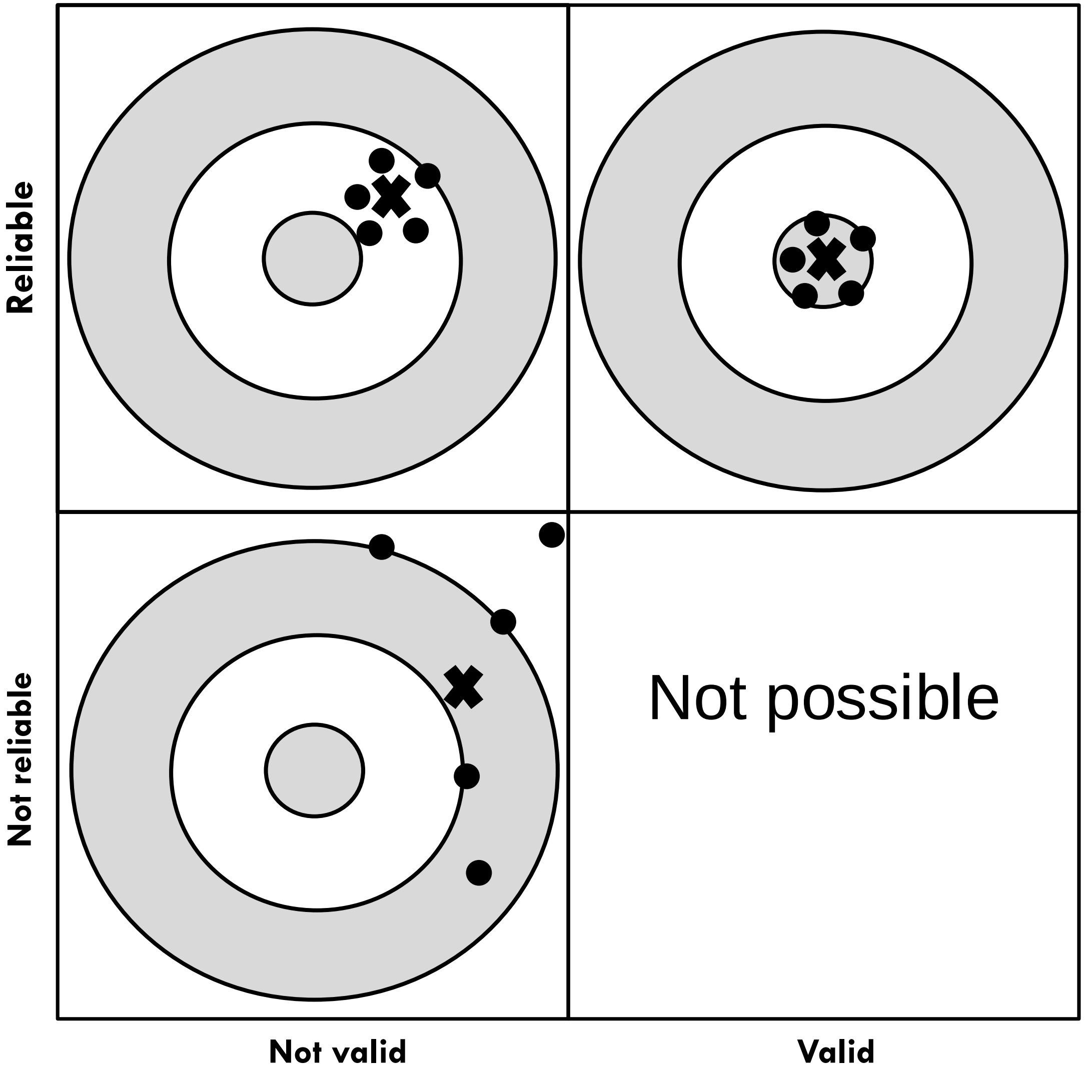
Reliability and Validity

True score model: $X_O = X_T + E_S + E_R$

- X_O = the observed score
- X_T = the true score
- E_S = systematic error
- E_R = random error

Validity refers to whether we are measuring what we want to measure (E_S=0)

Reliability is the degree to which what we measure is free from random error (E_R=0)



Type of Reliability

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- ***Test-Retest***
  - The same test over time
- ***Interrater***
  - Same test by different people
- ***Parallel forms***
  - Different versions of same test
- ***Internal Consistency***
  - Individual items of same test

Types of Validity

Construct Validity: correspondence between a measure at conceptual level and purported measure (includes the following types)

Face Validity

- Does the variable reflect what you want to measure? That is, "Does it makes sense?"

Content Validity

- Very similar to face validity, but more formalized, that is, questions used to measure a construct need to be closely related to the formal definition of the construct

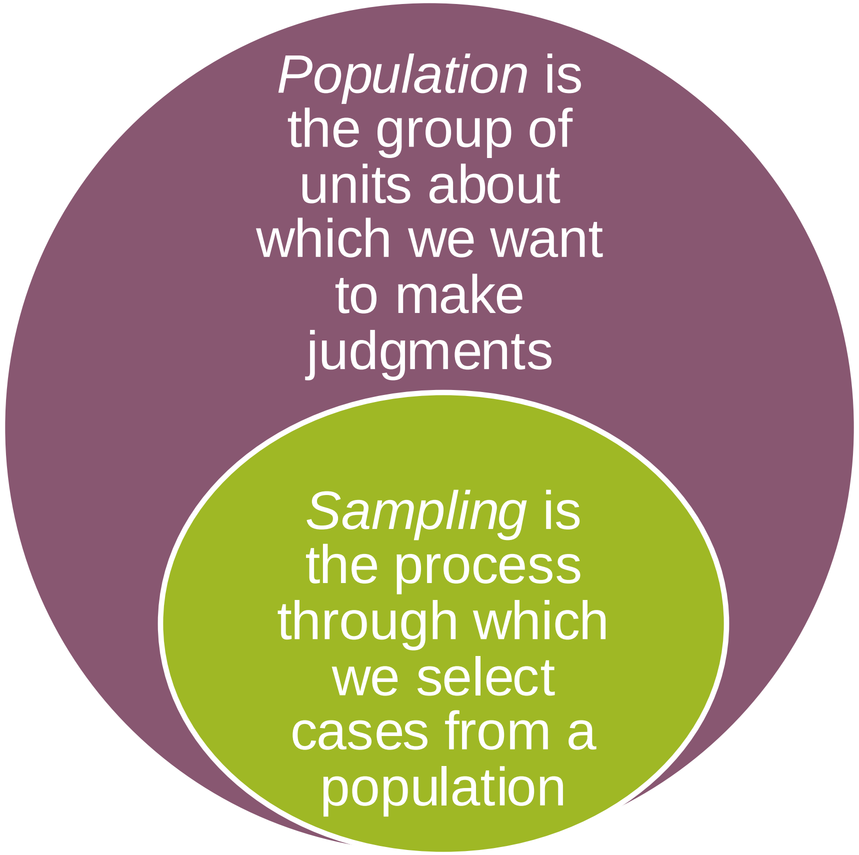
Predictive Validity

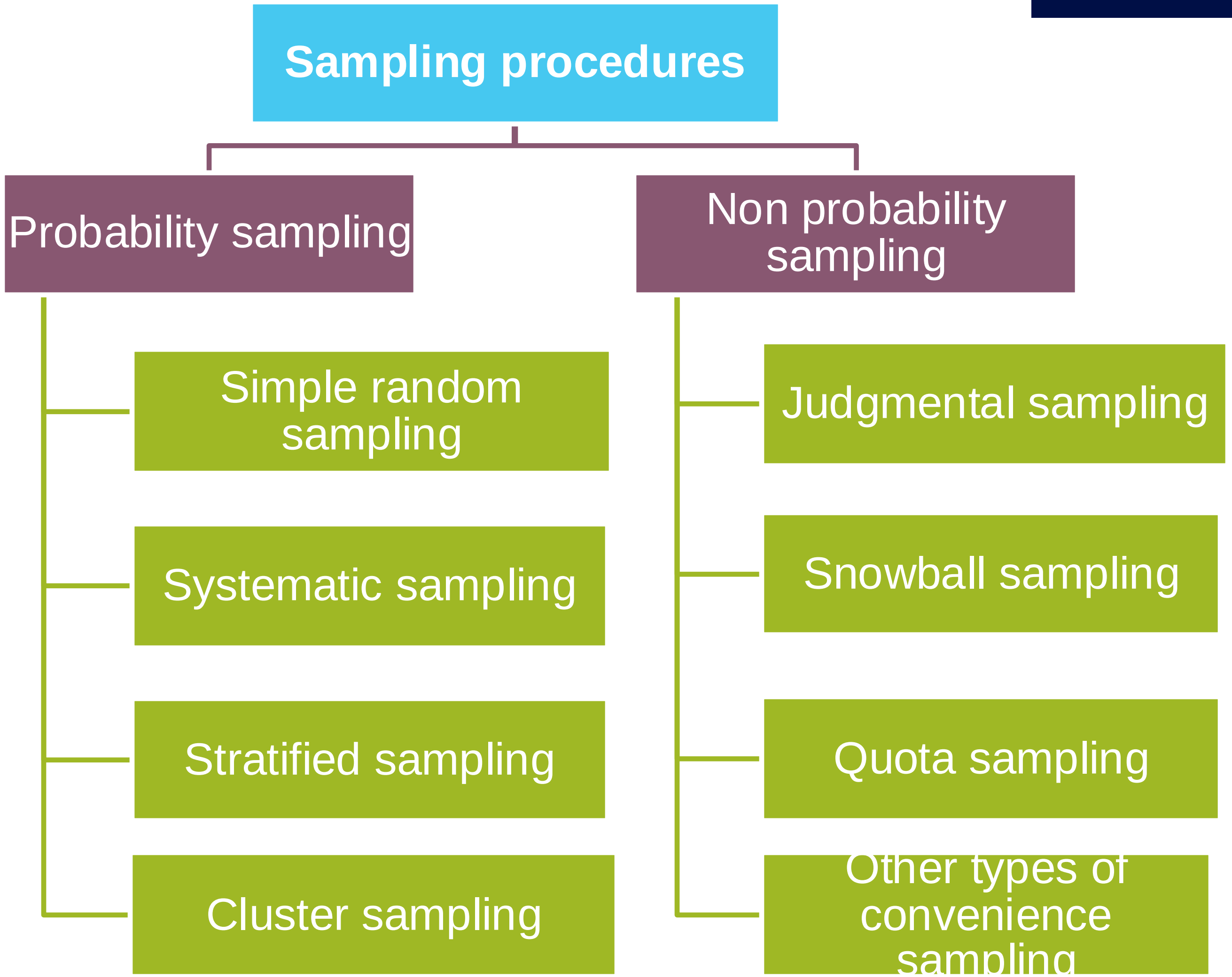
- Does the measure predict another phenomenon, measured in the future (e.g., satisfaction and future sales)?

Criterion Validity

- Like predictive validity, but measurement of the two phenomena takes place at the same time

Sampling





More to come in W6

Sample Sizes

Intuition: increasing sample size of a randomly selected sample helps reduce random error.

The marginal benefit of increasing the sample sizes becomes smaller as the sample becomes larger. An increase from n=40 to n=60 is more beneficial than from n=1,000 to n=1,020.

The sample size required is (almost) unrelated to the size of the population.

More to come in W6.

Takeaways

1. Secondary data can be collected by a range of internal and external sources for databases, articles, studies etc.

2. Nominal, ordinal, interval, and ratio scaled data contain different degrees of information.

3. Validity refers to the degree of systematic error; reliability to the degree of random error in measurement.

4. There are many types of sampling procedures, each with distinct benefits.